

Proof. By contradiction, assume that $A \neq B$. Then, without loss of generality, there exists $x \in X$ such that $x \in A$ and $x \notin B$. But then it is not true that $A \subseteq B$. The contradiction assumption must therefore be false, i.e. $A = B$. $\square$

Characteristic property

The elements of a subset $A \subseteq X$ can often be characterized by a mathematical property that they satisfy. This property is denoted $p(x)$, and we write

$$A = \{x \in X \mid p(x)\}.$$

For example, if $p(x) = 'x \text{ is even}'$, then

$$A = \{x \in \mathbb{N} \mid x \text{ is even}\} = \{0, 2, 4, \dots\} \subseteq \mathbb{N}.$$

Operations on sets

- **Complement:** if $A \subseteq X$ then we define its *complement* to be

$$A^C = \{x \in X \mid x \notin A\}.$$

- **Union:** for two sets $A \subseteq X$ and $B \subseteq X$ we define their *union* to be

$$A \cup B = \{x \in X \mid x \in A \text{ or } x \in B\}$$

- **Intersection:** for two sets $A \subseteq X$ and $B \subseteq X$ we define their *intersection* to be

$$A \cap B = \{x \in X \mid x \in A \text{ and } x \in B\}$$

- **Difference:** for two sets $A \subseteq X$ and $B \subseteq X$ we define their *difference* to be

$$A \setminus B = \{x \in X \mid x \in A \text{ and } x \notin B\}$$

- **Disjoint Union:** for two sets $A \subseteq X$ and $B \subseteq X$ whose intersection is empty, we often replace the symbol $\cup$ by

$$A \sqcup B$$

Lemma 2.2 (Properties of $\cap$ and $\cup$): For some set X and subsets $A, B, C \subseteq X$ the operations $\cap$ and $\cup$ satisfy:

1. Boolean properties: $A \cap A^C = \emptyset$ and $A \cup A^C = X$.
2. Commutativity: $A \cap B = B \cap A$ and $A \cup B = B \cup A$.
3. Associativity: $(A \cap B) \cap C = A \cap (B \cap C)$ and $(A \cup B) \cup C = A \cup (B \cup C)$.
4. Distributivity: $(A \cap B) \cup C = (A \cup C) \cap (B \cup C)$ and $(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$.

5. De Morgan laws: $(A \cap B)^C = A^C \cup B^C$ and $(A \cup B)^C = A^C \cap B^C$.

Proof. We prove the first of the De Morgan laws. The rest is an exercise.

We want to show that two sets are the same: $(A \cap B)^C = A^C \cup B^C$. To do this, we will show that the set on the left is contained in (or equal to) the set on the right, and vice versa. I.e., we shall show that $(A \cap B)^C \subseteq A^C \cup B^C$ and $(A \cap B)^C \supseteq A^C \cup B^C$.

(i) To show that $(A \cap B)^C \subseteq A^C \cup B^C$, we note the following implications:

$$\begin{aligned}
 x &\in (A \cap B)^C \\
 &\Downarrow \\
 x &\notin A \cap B = \{y \in X \mid y \in A \text{ and } y \in B\} \\
 &\Downarrow \\
 x &\notin A \text{ or } x \notin B \\
 &\Downarrow \\
 x &\in A^C \text{ or } x \in B^C.
 \end{aligned}$$

Since $A^C \subseteq A^C \cup B^C$ and $B^C \subseteq A^C \cup B^C$, we conclude that necessarily $x \in A^C \cup B^C$. Hence $(A \cap B)^C \subseteq A^C \cup B^C$.

(ii) Conversely, we can show $(A \cap B)^C \supseteq A^C \cup B^C$. Assume that $x \in A^C \cup B^C$ and *by contradiction*, assume that $x \notin (A \cap B)^C$. Then we have the implications:

$$\begin{aligned}
 x &\notin (A \cap B)^C \\
 &\Downarrow \\
 x &\in A \cap B = \{y \in X \mid y \in A \text{ and } y \in B\} \\
 &\Downarrow \\
 x &\in A \text{ and } x \in B \\
 &\Downarrow \\
 x &\notin A^C \text{ and } x \notin B^C.
 \end{aligned}$$

But this is in contradiction to the assumption that $x \in A^C \cup B^C$. Therefore the contradiction assumption $x \notin (A \cap B)^C$ is not true, hence $x \in (A \cap B)^C$.

We have shown that $(A \cap B)^C \subseteq A^C \cup B^C$ and that $(A \cap B)^C \supseteq A^C \cup B^C$, so by Lemma 2.1 the two sets must be equal, completing the proof. $\square$

2.2 Elements of mathematical logic

The building blocks of mathematical logic are **formulas**, which can be either *true* or *false*. Here are some examples:

$$\begin{aligned}
 p &= \text{'Blue is a color'} \\
 q &= \text{'15 is the square of a natural number'} \\
 r &= \text{'the number 3 belongs to the set } X'
 \end{aligned}$$

Then p is true, q is false, and we have no way of knowing whether r is true or false without knowing something about the set X .

2.2.1 Connectives

Connectives are the tools for building new formulas from existing ones. We briefly mention them:

Logical negation $\neg p$ ('not p ') is the negation of the formula p

Logical conjunction $p \wedge q$ (' p and q ')

Logical disjunction $p \vee q$ (' p or q ')

Logical implication $p \Rightarrow q$ (' p implies q ' or 'if p , then q ')

Logical equivalence $p \Leftrightarrow q$ (' p is logically equivalent to q ')

Proof by contradiction

This formalism allows us to understand the notion of a proof by contradiction, which is summed up by the logical equivalence:

$$(p \Rightarrow q) \quad \Leftrightarrow \quad (p \wedge \neg q \Rightarrow \neg p)$$

2.2.2 Predicates

A **predicate** is a formula that depends on one or more variables. In fact, we have seen predicates before, when we called them 'characteristic properties'. Here are some more examples:

$p(x)$ = ' x is a prime number'

$q(y)$ = ' y is the square of a natural number'

$r(x, y)$ = ' x is divisible by y '

2.2.3 Quantifiers

In a set X , for a given predicate $p(x)$ with $x \in X$, we can ask whether p is always true, or perhaps only sometimes. This is expressed mathematically as follows:

Universal quantifier: $\forall x, p(x)$ (we say 'for every x , $p(x)$ holds')

Existential quantifier: $\exists x, p(x)$ (we say 'there exists x , such that $p(x)$ holds')

Unique existential quantifier: $\exists!x, p(x)$ (we say 'there exists one and *only one* x , such that $p(x)$ holds')

Example 2.1: Suppose that, as above, $p(x)$ = ' x is a prime number'. If $X = \mathbb{N}$, then it is true that *there exists* $x \in X$ that is a prime number, i.e. $\exists x, p(x)$. However, it is *not true* that *every* $x \in X$ is a prime number. That is, $\neg(\forall x, p(x))$.

Example 2.2: Consider the predicate $p(x) = 'x^2 = x'$. If $X = \{1, 2, 3, \dots\}$, then $1 \in X$ is the unique element in X for which $p(x)$ is true. That is, $\exists!x, p(x)$. On the set $Y = \{2, 3, 4, \dots\}$ the predicate $p(x)$ is never true, i.e. $\neg(\exists x, p(x))$.